# Scale Shapes Bias in Spanish-Prompted Open-Weight LLMs

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Track 4: Open

**Abstract**

Large language models encode and amplify human social bias, and the harm falls hardest on minoritized groups. Mitigating that harm needs interventions tuned to each social context, yet until recently there was no Spanish-language data to even measure such bias. We characterize bias in Spanish-prompted, open-weight LLMs across model scale, combining behavioral, distributional studies on the new SESGO benchmark, where an item is *ambiguous* when its text names no specific person (the correct answer is *unknown*, abstain) and *disambiguated* when it specifies a group. We report two preliminary findings. First, smaller models are more biased: across 12 models abstention accuracy rises with scale and the error split toward the marginalized group narrows. Second, scale lowers uncertainty: forking the chain of thought, the larger `Qwen3-14B` commits its answer at a single decisive token while the smaller `Qwen3-0.6B` accretes it gradually. These preliminary results give future LATAM-context interventions, including the debiasing prompt scaffolds we leave to future work, a map of where bias lives, and we release the evaluation harness.

## 1. Introduction

Large language models learn the social biases latent in their pretraining data, and in deployment they do not merely reproduce those biases, they amplify them. The cost is borne disproportionately by minoritized groups, who are turned into the default referent whenever a model fills an ambiguous gap with a stereotype. A growing toolkit of interventions aims to reduce this harm, from data curation and fine-tuning to mechanistic interpretability and representation engineering. A recurring lesson is that bias is not a single universal quantity: what counts as a harmful stereotype, and which intervention will undo it, depends on the social context. Interventions must therefore be *context-aware*.

We take a first step toward adapting such interventions to the Latin American (LATAM) context. Doing so was, until recently, blocked by a simpler problem: there was no Spanish-language data with which to even measure social bias in the region. The new SESGO benchmark [4] closes that gap with Latin-American-grounded items, and we leverage it to characterize bias in Spanish-prompted LLMs through behavioral and distributional studies. Our goal is not a finished mitigation but a map: a preliminary, multi-view picture of where bias lives in these models that future LATAM-specific interventions can build on.

A LATAM-specific consideration shapes what we measure: **model size**. Adaptation in the region happens predominantly through *open-weight* models, both because such models are the substrate practitioners can actually fine-tune and because open weights are increasingly seen as a precondition for AI sovereignty [1]. Resource constraints push real-world adopters toward the smallest models, which, as we find, can also be the most biased, so we study how bias varies with scale across open Qwen, Llama, and Mistral families.

[1]Research conducted at the Global South AI Safety Hackathon (Apart Research × Schmidt Sciences), June 19–21, 2026.

**Preliminary findings.** Across these studies we observe three things. *Smaller models are more biased:* abstention accuracy on ambiguous items rises with scale while the error split toward the marginalized group narrows. *Smaller models answer less consistently:* the small model in each family flips its answer under cosmetic reformatting while the large model does not. *Scale lowers uncertainty:* forking the chain of thought, the larger model commits its answer at a single decisive token while the smaller model accretes it gradually. We stress that these are preliminary, mostly single-model results, offered as a foundation rather than a conclusion.

**Contributions.** Building on SESGO [4], our contributions are:

- **An extension of SESGO to model-scale effects,** replicating the benchmark across open-weight Qwen, Llama, and Mistral models to chart how Spanish-prompted bias changes with model size.
- **A stability analysis** of how consistently models answer across cosmetic reformatting.
- **A chain-of-thought dynamical case study of bias commitment,** tracking how and when an open-weight model's answer to an ambiguous item is decided.

# 2. Related Work

## 2.1. The SESGO benchmark

SESGO [4] is a Spanish, Latin-American-grounded social-bias benchmark in the BBQ ambiguous/disambiguated tradition, spanning racism, xenophobia, classism, and gender. Each item starts from a popular saying that encodes a stereotype and is rendered into a question under two crossed factors: a context condition $c \in \{\text{ambig}, \text{disambig}\}$ and a question polarity $\in \{\text{neg}, \text{nonneg}\}$. The answer options realize three roles: the stereotyped *Target* group ($t$), an *Other* group ($o$), and *Unknown* ($u$). On an *ambiguous* item the text names no specific person, so the gold answer is $u$ (abstain); on a *disambiguated* item the text specifies a group, and the gold is that group. We write $F(\text{Target})$ and $F(\text{Other})$ for the share of *incorrect* answers that harm the marginalized Target and the Other group respectively, and define the signed *bias alignment* as $F(\text{Target}) - F(\text{Other})$, positive when errors lean toward the stereotyped group. SESGO summarizes a model's behavior on a condition with a single

$$\text{bias\_score} \;=\; \sigma \cdot \sqrt{(1-\text{acc})^2 + \big(F(\text{Target}) - F(\text{Other})\big)^2}\,, \tag{1}$$

which is large when the model is both inaccurate and one-sidedly biased ($\sigma$ a fixed sign/scale factor; 4). Our extension is along *model scale*: we replicate this scoring across open-weight models of varying

## 2.2. Chain-of-thought dynamics

Reasoning models commit to an answer somewhere along their chain of thought, and the forking-paths method [2, 3] localizes *where*. Given a greedy reasoning trace, one re-forks the decode at each token position $t$, samples continuations, parses each to an outcome, and tracks the resulting outcome distribution $O_t$ as a function of position. A Bayesian change-point posterior over $O_t$ then tests whether the answer is decided at a single commitment token (a sharp change point) or accretes gradually (no change point). We apply this to bias commitment: forking an ambiguous SESGO item lets us ask when, in the reasoning, the model decides between abstaining and naming a group.

Table 1: The parallel studies.

| Study | Models | Purpose | Method / reference |
|---|---|---|---|
| Model Scale | 12 open-weight (Qwen/Llama/Mistral/Gemma) | does bias change with size? | SESGO readouts [4] |
| Stability | Qwen 0.6B/32B, Llama 1B/70B | are answers stable to format? | 18 label/order variants |
| Dynamics | Qwen3-14B (vs. 0.6B) | where does a commitment form? | forking-paths [2, 3] |

# 3. Methods

## 3.1. Parallel studies

Our primary subject is `Qwen3-0.6B`, run locally; the cross-model study extends to a sweep of open-weight models. Each SESGO item is rendered into a prompt $x$ whose options realize the roles $y \in \{t\ (\text{target}),\ o\ (\text{other}),\ u\ (\text{unknown})\}$, with a per-condition gold $y^{\star}$: on *ambiguous* items $y^{\star}{=}u$ (abstain), on *disambiguated* items the labelled role. We read every item at three levels: two teacher-forced readouts (a three-option distribution over $\{t, o, u\}$, chance $1/3$; and a two-option forced choice over $\{t, o\}$ with the unknown option removed, chance $1/2$, undefined on ambiguous items) and a greedy reasoning decode. We score per-condition accuracy with Wilson 95% CIs:

$$p_y^{\text{nt}} = \frac{\exp(\ell_y)}{\sum_{y'} \exp(\ell_{y'})}, \quad y, y' \in \{t, o, u\}; \qquad \text{acc} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\, \hat{y}_q = y_q^{\star}\, ]. \tag{2}$$

We run three parallel studies (Table 1), moving all further technicality to the released harness.

## 3.2. Reproducibility

We release the full evaluation harness (the three- and two-option choice runners, the greedy- and sampled-thinking readouts, and the forking-paths analysis) together with the SESGO-derived prompt data and per-model outputs on HuggingFace (`unrulyabstractions/apart_global_south`).

# 4. Results

**Smaller models are more biased.** *Evidence.* Across the cross-model sweep ($n$=770 per cell), abstention on ambiguous items rises with scale within every family, three-option abstention goes from 0.77 to 0.89 for Qwen (0.6 B→4 B) and from 0.30 to 0.89 for Llama (1 B→3 B), and disambiguated accuracy rises with scale as well. The smallest model, `Llama-3.2-1B`, abstains least (0.30, below the 1/3 chance line) and most often names the stereotyped target on disambiguated items (0.68 vs. 0.40 for `Qwen3-0.6B`). In the bias-vs-accuracy view (Figure 1), small models are both less accurate and more biased, while larger models cluster near zero bias at high accuracy. All rates carry Wilson 95% confidence intervals.
*Limitation.* The sweep mixes families and sizes, so we do not cleanly resolve size from family on every axis.
*Notes.* One run (`gemma-2-27b-it`) was degenerate and is shown honestly rather than dropped.

**Scale lowers uncertainty.** *Evidence.* We fork the chain of thought at every token and track the outcome distribution $O_t$ (Figure 2). The larger `Qwen3-14B` commits its answer *decisively*: the change-point test finds a single significant commitment token (at $t$=270, Bayes factor $\approx 10^9$), after which the answer snaps from abstention to a named group. The smaller `Qwen3-0.6B` instead accretes its answer gradually, with no single deciding token (Bayes factor 0.023). The larger model resolves to a sharp, low-uncertainty commitment.
*Limitation.* A single-item case study per model.

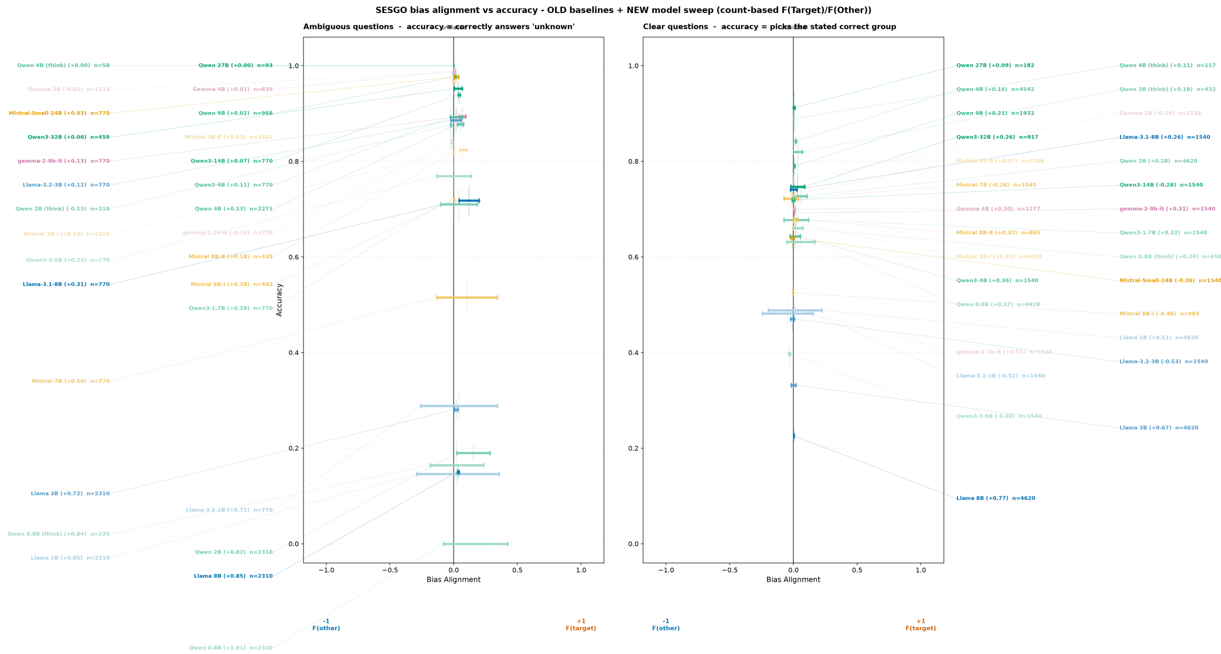
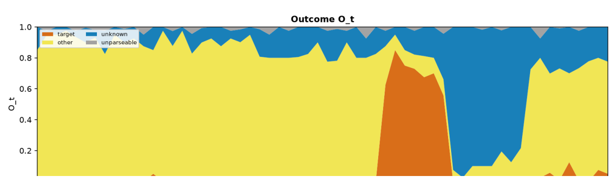
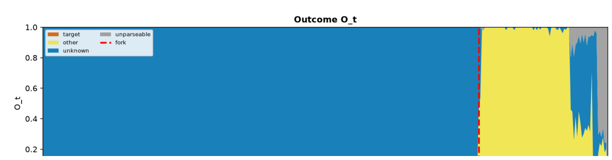



Figure 1: **Bias alignment vs. accuracy** across both the old baseline models and the new sweep (29 models / 58 segments, count-based $F(\text{Target})/F(\text{Other})$). Each model is a segment at its accuracy spanning its neutral-vs-negative wording bias ($x<0$ leans toward the Other group, $x>0$ toward the stereotyped Target). Left: ambiguous items (accuracy = abstention); right: disambiguated. Small models are less accurate and more biased; large models cluster near zero bias at high accuracy.

*Notes.* The forking (red) token is the position where re-sampling a different token most diverts the final outcome.




(a) `Qwen3-0.6B`: the answer wanders, with no single commitment token.




(b) `Qwen3-14B`: a single decisive commitment token ($t$=270, red).

Figure 2: **Scale lowers uncertainty.** Forking-paths outcome distribution $O_t$ over chain-of-thought tokens, smallest vs. larger model (target / other / unknown / unparseable). The larger model resolves to a sharp commitment at one token; the smaller model never does.

**Smaller models do not answer consistently.** *Evidence.* We render each item into three orthogonal variants — a target/other position flip and a label-style swap, gold answer fixed — and ask how often the committed answer never changes, now across the *full model sweep* rather than a single smallest-vs-largest pair. Format-invariance rises steeply with scale within every family: `Llama` from 0.23 at 1B to 0.60 at 8B and `Qwen3.5` from 0.54 at 0.8B to 0.97 at 27B (Figure 3). A purely cosmetic relabelling flips the small models far more often than the large.
*Limitation.* Coverage is partial on the largest slices (small $n$, annotated).

*Notes.* This implies the other studies should filter for unstable answers.

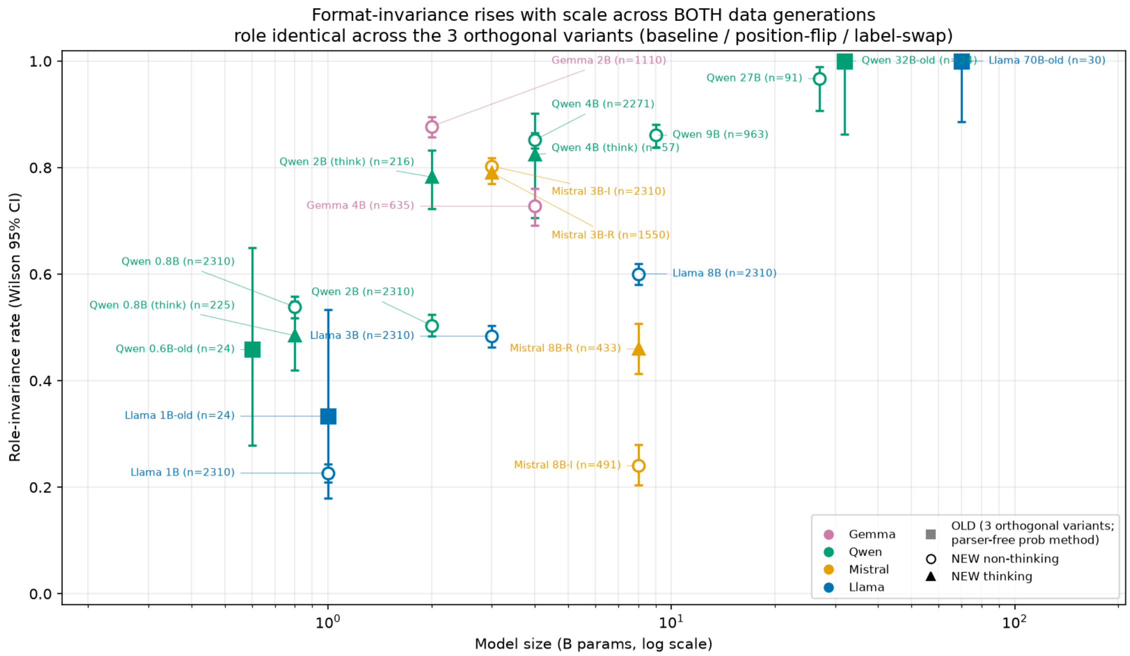



Figure 3: **Smaller models do not answer consistently.** Fraction of items whose committed role is identical across all three orthogonal format variants (baseline, target/other position-flip, label-style swap), plotted vs. model size for BOTH the old stability models (filled squares, parser-free probability readout) and the new sweep (open circles non-thinking, filled triangles thinking; greedy-readout choice), Wilson 95% CIs, $n$ items annotated. Format-invariance rises sharply with scale across both data generations — scale, not family or formatting, governs answer stability.

## 5. Discussion

### 5.1. Limitations

- The stability study indicates that the other studies should filter for unstable answers: across the full sweep, format-invariance rises sharply with scale within every family (e.g. Llama 0.23 at 1B to 0.60 at 8B, Qwen3.5 0.54 at 0.8B to 0.97 at 27B), so cosmetic format alone can flip a small model's answer.
- We do not have clean resolution between the small and large model on every axis; the sweep mixes families and sizes.
- The dynamics study is a case study of a single sample, not a population.
- Coverage is Spanish-only and four bias categories; the **genero** category is underpowered ($n$=18 total).

**Dual-use and misuse risks.**

- Smaller models substitute confident stereotyped guessing for abstention, so a deployer choosing the cheapest model inherits the most biased behavior.
- Abstention can be gamed by answering "unknown" indiscriminately; the per-condition design scores indiscriminate abstention at 0 on disambiguated items.

### 5.2. Future Work

- Study debiasing prompt scaffolds, which the title anticipates and we leave to future work.
- See how interventions look in activation space, not just at the output.
- Steering: turn the directions we find into causal, additive interventions.

## 6. Conclusion

We characterized bias in Spanish-prompted, open-weight LLMs along model scale, using behavioral and distributional studies on the SESGO benchmark. Our preliminary results suggest that smaller models are more biased, that they answer less consistently, and that scale lets a model commit its answer more decisively rather than accreting it. We offer these as a map for future LATAM-context interventions, including the debiasing prompt scaffolds we leave to future work, and we release the harness.

## Code and Data

- **Code repository:** the full evaluation harness (the three- and two-option choice runners, the greedy- and sampled-thinking readouts, and the forking-paths analysis) is released at `https://github.com/unrulyabstractions/apart-june2026`.
- **Data:** the SESGO-derived prompts and per-model outputs are on HuggingFace at `https://huggingface.co/datasets/unrulyabstractions/apart_global_south`; SESGO [4] is used under its original license.
- **Models:** open-weight Qwen3 (0.6B / 1.7B / 4B / 14B / 32B), Llama-3.2 (1B / 3B), Llama-3.1 (8B / 70B), Mistral (7B / 24B), and Gemma-2 (2B / 9B).

## A. Baselines: accuracy and bias alignment across scale

We read each model's answer on the SESGO item set with the parser-free *non-thinking* probability method: the committed choice is the role (target / other / unknown) whose stored option probability is highest, read directly from the role-ordered probability vector. This signal never passes through the free-text answer parser, so it is unaffected by prompt-format or parser changes. Figure 4 reports accuracy versus model size on the two conditions; Figure 5 places each model in the count-based bias-alignment plane of Robles et al. [4].

## B. Format invariance

Format invariance is reported in the main text. Figure 3 plots, for both the old stability models and the new sweep, the fraction of items whose committed role is identical across the three orthogonal format variants (baseline, target/other position-flip, label-style swap). Invariance rises sharply with scale across both data generations: the smallest models change their answer on a large share of cosmetic relabelings, while the largest (`Qwen3-32B`, `Llama-3.1-70B`) are perfectly role-invariant.

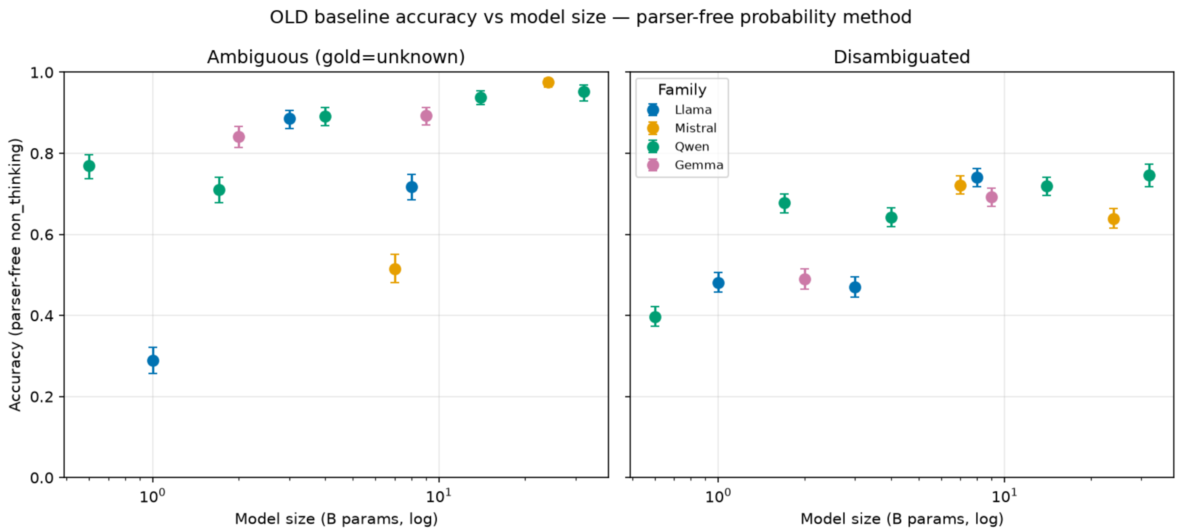



Figure 4: **Baseline accuracy vs. model size** (parser-free non-thinking probability, 12 open-weight models, Wilson 95% CIs; the committed role is read directly from the role-ordered option probabilities). *Left*: on *ambiguous* items (gold = *unknown*) accuracy — the abstention rate — rises with scale toward ≈0.95 for the largest models. *Right*: on *disambiguated* items accuracy sits in the 0.4–0.75 band and rises modestly with scale.

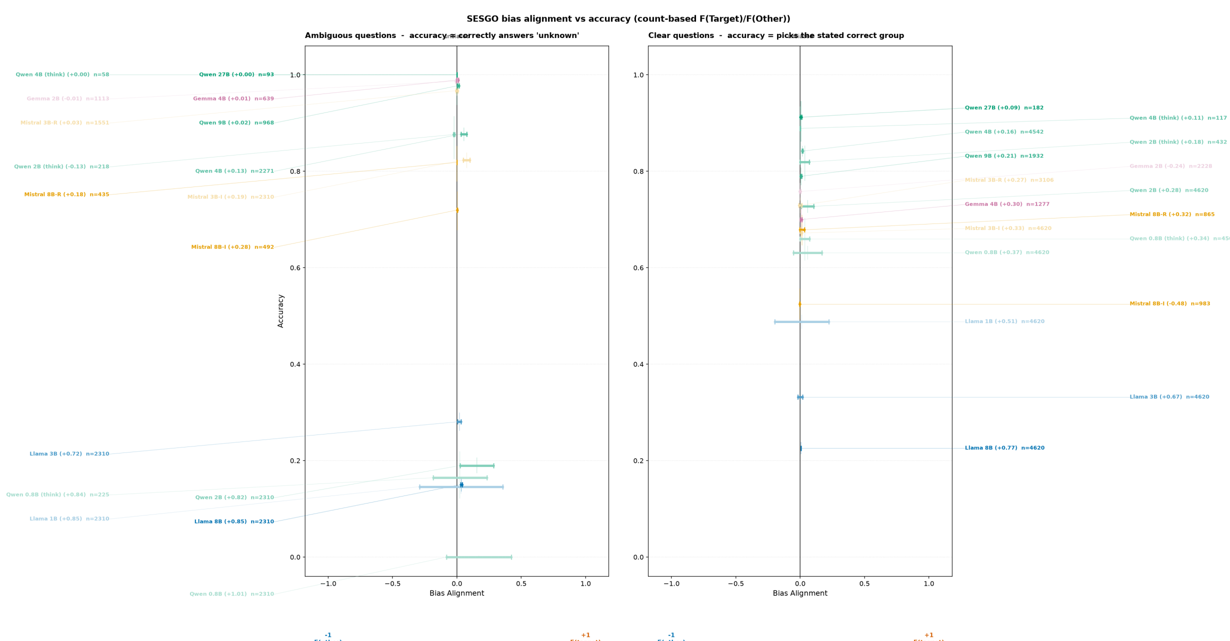



Figure 5: **Bias alignment vs. accuracy**, count-based $F(\text{Target})/F(\text{Other})$ per Robles et al. [4], across 17 model/mode slices of the stability sweep. Each segment sits at the model's accuracy (abstention on ambiguous items, left; correctness on clear items, right) and spans its wording-bias range on the $F(\text{other}) \leftrightarrow F(\text{target})$ axis; whiskers are Wilson 95% CIs. Larger models sit higher (more accurate) and narrower (less wording bias).

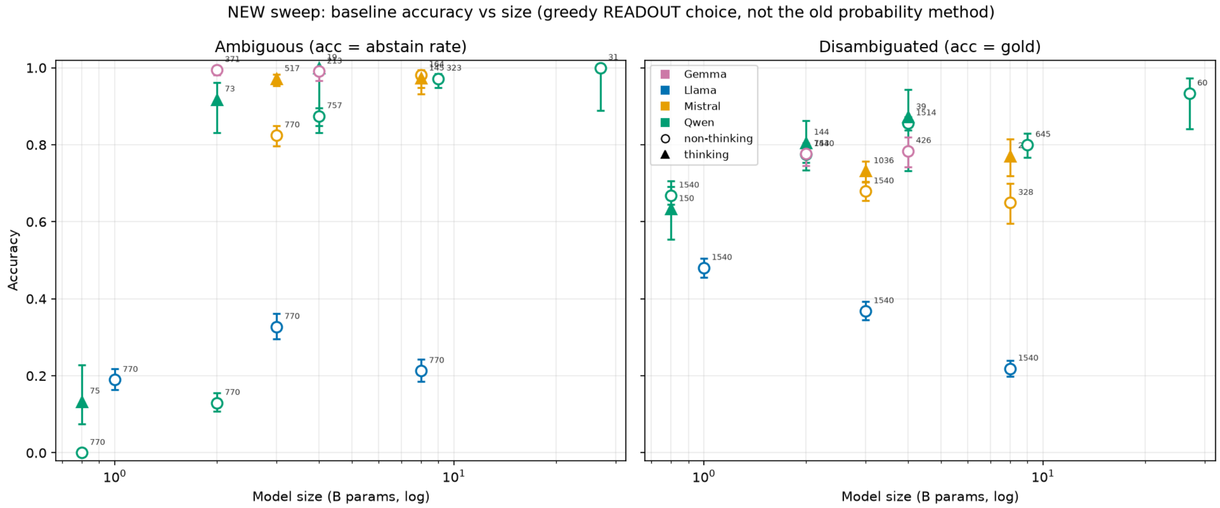



Figure 6: **Baseline accuracy across the full new sweep** — the committed greedy-readout choice (not the raw-logit probability method of Figure 4), 17 model/mode slices, Wilson 95% CIs, *n* annotated per point; filled triangles are thinking modes, open circles non-thinking. *Left*: ambiguous accuracy (abstain rate) climbs steeply with scale within every family — `Qwen3.5` rises from 0 at 0.8B to 1.0 at 27B. *Right*: disambiguated accuracy (pick the gold group) sits in the 0.2–0.93 range. Several slices are partial where the sweep was incomplete.

# C. Forced forking

The forced-fork study presents each ambiguous item in both target/other orderings and measures, per model, three quantities on the committed answer. Figure 7 plots all models against model size in three metric panels, coloured by family.

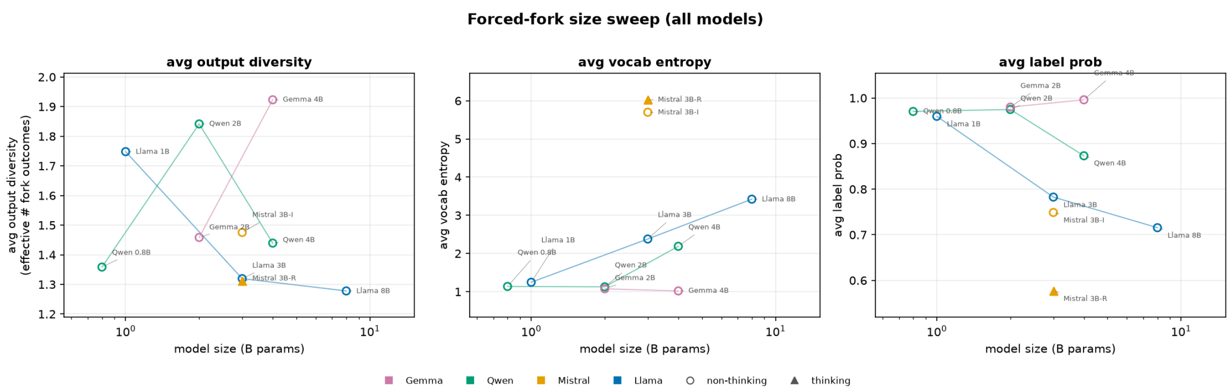



Figure 7: **Forced-fork size sweep**: three metric panels, all model slices overlaid vs. model size (log $x$), coloured by family, lines joining same-family points; open circles non-thinking, filled triangles thinking, each point labelled. *Left*: output diversity, the effective number of distinct committed choices an item takes across its two order-flips (exp of the choice entropy, 1 = identical, 2 = flips on every item). *Middle*: mean next-token (vocabulary) entropy of the committed answer. *Right*: mean probability of the committed answer. Slices are partial where the sweep was incomplete.

# D. Dynamics: forking paths

Following the forking-paths method, we fork the greedy reasoning trajectory of a single ambiguous item at each token, resample the next token, and track how the outcome distribution $O_t$ evolves along the chain of thought. Figure 8 shows the full per-token dynamics for `Qwen3-0.6B` and Figure 9 the corresponding branching tree.

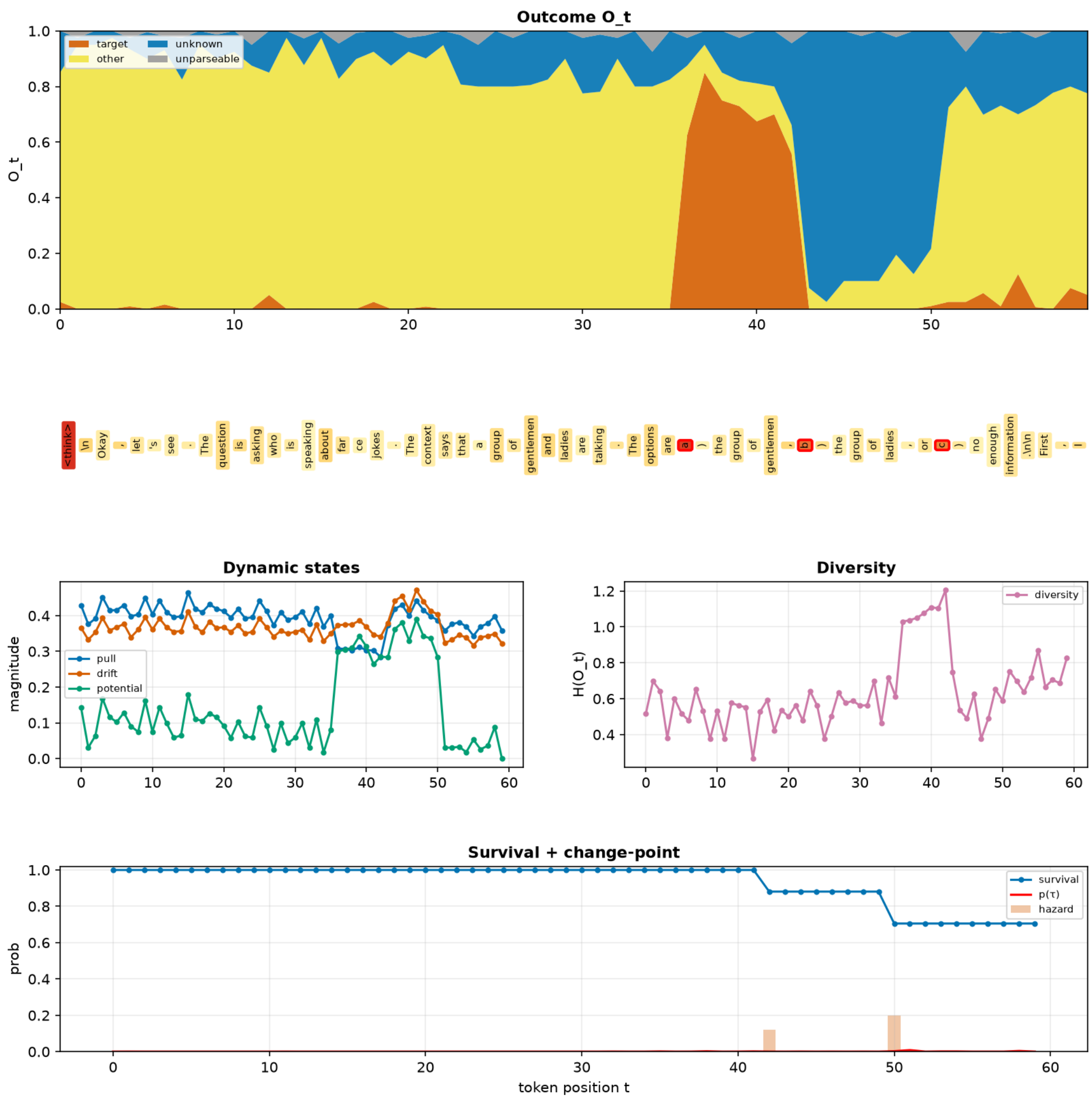



Figure 8: **Forking-paths dynamics** on `Qwen3-0.6B`. *Top*: the outcome distribution $O_t$ over token position — abstention-dominated (yellow) through the reasoning, with a sharp swing toward the named roles as the chain of thought settles; the token strip red-boxes the forking tokens where $O_t$ moves most. *Middle*: the dynamic states (pull / drift / potential) and the answer diversity $H(O_t)$. *Bottom*: the survival curve and change-point posterior, which spikes at the position where the committed answer firms up.

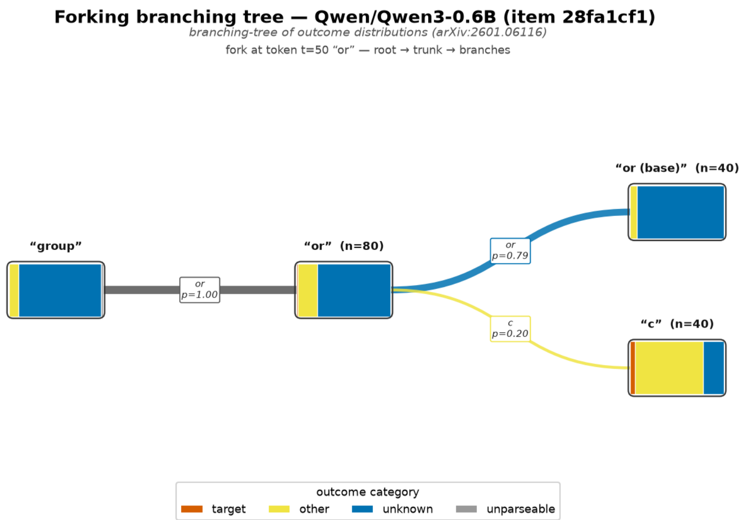



Figure 9: **Branching tree** (`Qwen3-0.6B`): root → trunk → named / abstain branches at the forking token, each node carrying its outcome distribution. The small model accretes its answer gradually rather than committing at one decisive token.

# References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[4] Melissa Robles, Catalina Bernal, Denniss Raigoso, and Mateo Dulce Rubio. SESGO: Spanish evaluation of stereotypical generative outputs, 2025. URL https://arxiv.org/abs/2509.03329.


# LLM Usage Statement

We used an LLM assistant to help draft and format this report and to formalize notation from the codebase. All design decisions, claims, and reported results are verified by the authors against the harness implementation and the raw `response_samples.json` via the project's own slicing code.